

ResQAI: AI-Assisted Humanitarian Disaster Message Classification

Project Proposal
Submitted to:

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August 26, 2026

1 Higher-order goal

... secondary framing

This project aims to improve the efficiency of disaster information processing by automatically identifying and organizing humanitarian information from large volumes of social-media messages.

During natural disasters and humanitarian emergencies, affected individuals often use social media to report injuries, request assistance, share information about damaged infrastructure, communicate evacuation requirements, report missing people, or offer rescue and volunteering support.

The higher-order goal of ResQAI is to transform this unstructured and high-volume information into structured, actionable categories that can support emergency-response personnel and other decision makers.

The immediate engineering objective is to develop a deployable software pipeline that can classify disaster-related messages rapidly and provide additional contextual analysis for ambiguous messages.

2 Time-to-value

... primary heuristic

- We will deliver meaningful increments quickly by first implementing the core user workflow and validating the classification component.
- We will prioritize a working end-to-end prototype over unnecessary infrastructure, allowing the core classification workflow to be demonstrated early.
- The first iteration will focus on message input, humanitarian classification, confidence estimation, and result presentation.
- Subsequent iterations will add contextual analysis, structured information extraction, and improved decision-support functionality.
- Success will be measured using classification accuracy, weighted F1 score, inference behavior, and the usefulness of the resulting structured information.

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3 Problem Statement

Natural disasters generate large amounts of social-media communication in a short period of time. These messages may contain valuable information about people, infrastructure, emergency needs, evacuation, rescue activities, and safety conditions.

However, several challenges make it difficult to use this information effectively:

- **Information overload:** A large number of messages may be generated during a disaster, making manual inspection difficult.
- **Unstructured information:** Messages are generally short, informal, and lack a standardized structure.
- **Mixed relevance:** Useful humanitarian information is mixed with political discussions, opinions, emotional reactions, news, and unrelated content.
- **Semantic ambiguity:** Several humanitarian categories can have overlapping meanings, making classification difficult.
- **Limited response time:** Emergency information may become less useful if it cannot be identified quickly.

Why this matters now (contextual relevance): Social-media platforms can provide near-real-time information during disasters, but the usefulness of this information depends on the ability to filter, organize, and interpret it efficiently.

4 Proposed Solution

... Software Proposition

4.1 Overview

ResQAI is proposed as an AI-assisted disaster information processing system that automatically classifies incoming crisis-related messages into humanitarian categories.

The system consists of the following major components:

- **Message input:** Accept a disaster-related social-media message.
- **BERTweet classifier:** Use a transformer model pretrained specifically for Twitter-style text and fine-tuned on humanitarian disaster messages.
- **Confidence estimation:** Produce the predicted category together with a confidence score.
- **LLM fallback:** Route ambiguous or low-confidence messages to a large language model such as Gemini for additional contextual analysis.
- **Structured output:** Present the resulting category and, where applicable, additional information such as urgency, location, affected people, and recommended action.

4.2 Core workflow

1. A disaster-related message is submitted to ResQAI.
2. The message is tokenized using the BERTweet tokenizer.
3. The fine-tuned BERTweet classifier predicts one of the ten humanitarian categories.
4. The system obtains the prediction confidence.

5. High-confidence predictions are accepted directly.
6. Low-confidence predictions can be passed to the secondary Gemini analysis layer.
7. The system presents the classification and relevant structured information to the user.

4.3 Humanitarian classification categories

The implemented classifier uses ten categories:

- `caution_and_advice`
- `displaced_people_and_evacuations`
- `infrastructure_and_utility_damage`
- `injured_or_dead_people`
- `missing_or_found_people`
- `not_humanitarian`
- `other_relevant_information`
- `requests_or_urgent_needs`
- `rescue_volunteering_or_donation_effort`
- `sympathy_and_support`

5 Solution Approach

... Engineering focus

This is an engineering-oriented machine learning project focused on developing a practical system for disaster information classification.

The project uses established machine learning and natural language processing techniques rather than proposing a new transformer architecture.

5.1 Dataset and preprocessing

... data preparation

The project uses the HumAID humanitarian information dataset.

The dataset contains:

- 53,531 training messages,
- 7,793 validation messages,
- 15,160 test messages,
- 76,484 messages in total,
- 10 humanitarian classification categories.

Messages are tokenized using the BERTweet tokenizer.

The maximum sequence length is set to 128 tokens. Dataset analysis showed that the average message length is approximately 35 tokens, while only a very small proportion of messages reaches the 128-token limit.

5.2 Model development

... experimental engineering

Multiple approaches were evaluated before selecting the final model.

The investigated approaches included:

- TF-IDF with Naive Bayes,
- TF-IDF with Logistic Regression,
- class-weighted Logistic Regression,
- Linear SVM,
- Random Forest,
- DistilBERT,
- class-weighted DistilBERT,
- BERTweet.

BERTweet produced the strongest overall performance and was therefore selected as the primary classifier.

5.3 BERTweet fine-tuning

... primary ML component

The final BERTweet configuration uses:

- Learning rate: 1×10^{-5}
- Maximum sequence length: 128
- Training batch size: 8
- Evaluation batch size: 8
- Maximum epochs: 3
- Weight decay: 0.01
- Warm-up steps: 500
- Linear learning-rate scheduler
- FP16 mixed precision during training
- Early stopping
- Best validation weighted F1 checkpoint selection

5.4 Overfitting control

... generalization

The model is evaluated after each epoch using validation data.

The best-performing checkpoint is selected using weighted F1 rather than automatically using the final training epoch.

The experiments showed that validation performance peaked around the second epoch and decreased slightly during the third epoch. The use of early stopping and best-checkpoint selection therefore helps prevent unnecessary additional training from reducing generalization.

5.5 Confidence-based secondary analysis

BERTweet is used as the primary classifier.

For messages where the model is sufficiently confident, the classification result can be returned directly.

For ambiguous messages, a secondary LLM layer using Gemini is proposed.

The secondary layer can analyze:

- humanitarian category,
- urgency,
- location,
- number of people affected,
- requested assistance,
- recommended action.

The LLM therefore complements rather than replaces the trained domain-specific classifier.

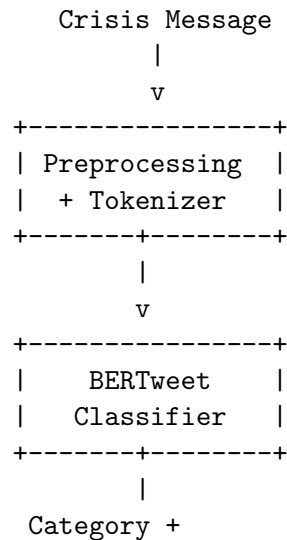
6 System Architecture

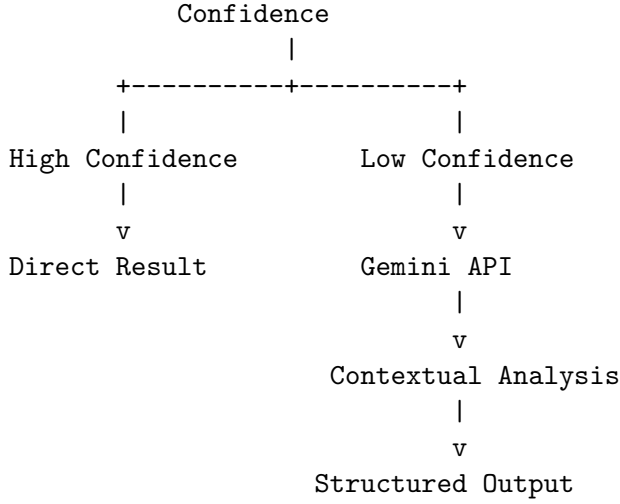
... software-based solution

The proposed ResQAI architecture consists of several logical components.

- **Input layer:** Receives a social-media or crisis message.
- **NLP layer:** Tokenizes and preprocesses the incoming message.
- **BERTweet layer:** Performs humanitarian classification.
- **Confidence layer:** Determines whether the classifier is sufficiently confident in its prediction.
- **LLM layer:** Provides additional analysis for ambiguous messages.
- **Output layer:** Displays classification, confidence, and additional structured information.

The conceptual architecture is:





7 Evaluation Criterion

... measurable and attributable

The success of the project will be evaluated using measurable machine learning and system-level metrics.

7.1 Primary evaluation metrics

Classification Accuracy:

$$Accuracy = \frac{CorrectPredictions}{TotalPredictions} \quad (1)$$

Accuracy measures the overall proportion of correctly classified messages.

Weighted F1 Score:

The F1 score combines precision and recall.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (2)$$

The weighted average accounts for the different support of individual classes.

7.2 Current achieved performance

The final BERTweet model achieved:

- **Best Validation Accuracy: 78.40%**
- **Best Validation Weighted F1: approximately 78.05%**
- **Test Accuracy: 78.67%**
- **Test Weighted F1: 78.27%**

The test results were obtained from the unseen test set containing 15,160 messages.

7.3 Secondary evaluation metrics

The following metrics can be considered during further system development:

- **Per-class precision and recall:** To identify categories that require improvement.

- **Confusion analysis:** To identify commonly confused categories.
- **Inference time:** Time required to classify a single message.
- **LLM fallback rate:** Percentage of messages requiring secondary analysis.
- **Confidence calibration:** Relationship between model confidence and actual prediction correctness.

7.4 Validation plan

...high-level

The model will be evaluated using the predefined training, validation, and test splits.

The validation set is used for model selection and hyperparameter decisions, while the test set is reserved for final performance evaluation.

This separation prevents the test set from influencing model selection.

8 Scalability

...with foresight

The proposition should scale in both theoretical and practical senses.

1. Scalable classification:

The trained classifier can process incoming messages independently, allowing multiple inference requests to be handled without retraining the model.

2. Batch processing:

Messages can be grouped into batches for more efficient processing when large volumes of information are available.

3. Separation of components:

The classifier, LLM analysis layer, and user interface can be separated into independent components.

4. API-based architecture:

The classification model can be exposed through an API so that different interfaces or applications can use the same inference service.

5. LLM cost control:

A confidence-based fallback mechanism avoids sending every message to an external LLM, which can reduce unnecessary API usage.

9 Engine availability heuristic

A mature set of machine learning and software components is available to implement the proposed system:

- **Hugging Face Transformers:** Model loading, tokenization, and transformer inference.
- **PyTorch:** Model execution and machine learning operations.
- **BERTweet:** Primary humanitarian text-classification model.
- **Gemini API:** Proposed secondary contextual reasoning layer.
- **Python:** Core implementation and integration language.

- **Web/API framework:** Can be used to expose the model as an application service.
- **Cloud or local deployment:** The inference component can be deployed on suitable compute infrastructure depending on the required scale.

The project uses established tools and models so that the engineering effort can focus on system integration, model evaluation, and practical usability.

10 Project scope and deliverables

... iteration-friendly

10.1 Initial deliverable

... first iteration

The initial implementation includes:

- HumAID dataset preparation.
- Training and evaluation of multiple baseline classifiers.
- Fine-tuning of BERTweet for ten-class humanitarian classification.
- Validation and test evaluation.
- Prediction of humanitarian category for a new message.
- Confidence score generation.
- Model checkpoint preservation for inference.

10.2 Subsequent deliverables

Further iterations will include:

- User-facing message input interface.
- Confidence-based Gemini fallback.
- Structured output generation.
- Urgency and priority identification.
- Location extraction.
- Recommended-action generation.
- Visualization of classification results.
- Optional storage of analyzed messages and results.

11 Risks and mitigations

- **Classification errors:**

The classifier may incorrectly categorize ambiguous messages.

Mitigation: Use confidence scores, detailed validation, confusion analysis, and secondary LLM analysis for low-confidence cases.

- **Semantic overlap between classes:**

Some humanitarian categories have inherently similar meanings.

Mitigation: Analyze confusion patterns and consider hierarchical or multilabel classification in future versions.

- **LLM dependency:**

The Gemini fallback requires external API access.

Mitigation: Keep BERTweet as the primary classifier and use the LLM only when required.

- **Incorrect emergency interpretation:**

Automated predictions should not directly trigger critical emergency actions without verification.

Mitigation: Treat the system as decision support and maintain human oversight for high-impact decisions.

- **Data privacy:**

Social-media messages may contain sensitive information.

Mitigation: Minimize stored information, avoid unnecessary personally identifiable information, and apply appropriate access controls.

- **Computational requirements:**

Transformer inference requires more resources than simple TF-IDF classifiers.

Mitigation: Use batching, model optimization, and appropriate deployment infrastructure when scaling.

12 Expected outcome

The expected outcome of the project is a working AI-assisted prototype that can accept a disaster-related message and automatically determine its most likely humanitarian category.

The primary machine learning component has already demonstrated a test accuracy of approximately 78.67% and a weighted F1 score of approximately 78.27%.

The completed system is expected to provide:

- automated humanitarian classification,
- confidence-based prediction,
- identification of ambiguous messages,
- optional LLM-based contextual analysis,
- structured disaster information,
- and a foundation for future real-time disaster information processing.

13 Summary

ResQAI proposes an AI-assisted approach for organizing and interpreting large volumes of disaster-related social-media messages.

The project focuses on time-to-value by first developing and validating the core classification model before adding more advanced system components.

The primary model, BERTweet, was fine-tuned using the HumAID dataset and achieved 78.67% accuracy and 78.27% weighted F1 on the unseen test set.

The system is designed around a practical hybrid architecture: BERTweet provides fast domain-specific classification, while a large language model such as Gemini can be used for additional contextual reasoning when the classifier is uncertain.

The project therefore combines machine learning, natural language processing, and software-system integration to create a practical information-processing tool for disaster-response scenarios.

The final objective is not to replace emergency responders but to reduce the amount of unstructured information they need to manually process and help surface potentially relevant humanitarian information more efficiently.